

Miranda Murarik

mirandamurarik@gmail.com | [linkedin.com/in/miranda-murarik](https://www.linkedin.com/in/miranda-murarik) | [mirandamurarik.com](https://www.mirandamurarik.com)

Eight years across research, analytics, data engineering, and AI engineering. Builds pipelines and LLM systems for the teams that depend on them, checks them against human baselines before they carry decisions, and hands them off so those teams run them.

WORK EXPERIENCE

TSIA **Jun. 2021 – Present**

AI Automation Engineer *Feb. 2026 – Present*

- Started TSIA's internal AI automation initiative across research and content operations, self-initiated and expanded to every process that fit, each built with the validation harnesses, reusable skills, and documentation the team needed to run it.
- Designed and shipped the LLM judging system behind an industry awards program, scoring 131 submissions in the 2026 live cycle. Reason-then-rate architecture across three data-derived personas on hybrid text and rendered page images, with independent models from two other providers as referees.
- Validated it against the human expert panel before it was trusted with decisions: matched the human winner in 7 of 7 categories across 29 held-out applications, with Impact reaching rho 0.71 against a human-to-human ceiling of 0.53. Established that ceiling first, diagnosing an apparent ICC of 0.14 to 0.24 as range restriction.
- Cut judging effort from 330 human grading hours to 10 hours of human-in-the-loop review, a 97% reduction.
- Replaced a 40-step manual poll checklist spanning five systems with a four-phase tool gated by four approvals, 1,246 lines across 16 Python modules. A first-time operator drove the second full cycle end to end with zero manual system edits, returning at least 32 hours a year per analyst.
- Deployed under real governance constraints. The AI pair-programmer never saw member data, working from structure and aggregate diagnostics only, verified against synthetic fixtures under read-only credentials.
- Won TSIA's company-wide AI Innovation and Enablement Award.

Principal Research Analyst *Feb. 2025 – Feb. 2026*

- Senior individual contributor for the research organization, owning solution design across the benchmark, survey, and conference programs.
- Replaced roughly 8 near-duplicate validation scripts with a single config-driven engine covering all 8 research practices, where adding a practice became a configuration entry instead of new code.
- Built the conference-to-CMS integration: 72 sessions filtered from 110 raw records, with author mapping validated against archived sessions at 58 of 61 exact, every mismatch explained, before any production write.

Data Engineer *Mar. 2022 – Feb. 2025*

- Led the migration of the firm's financial indices off manual vendor-workbook entry onto an API-driven Alteryx pipeline into BigQuery, with Tableau outlier detection. Owned the quarterly refresh from trainee to pipeline owner, 2022 through 2026.

Data Analyst *Jun. 2021 – Feb. 2022*

- Gathered requirements and mapped processes with stakeholders across teams. Delivered ad hoc statistical analysis, including a logistic regression on member renewal rates reported as non-significant ($p = .128$) rather than dressed up as a finding.

Azusa Pacific University **Jun. 2018 – May 2021**

Data Analyst, Grant Research, Dept. of Psychology *Azusa, CA*

- Ran point independently with senior stakeholders on five grant-funded projects across four partners: SAMHSA, the California Departments of Mental Health and Health Care Services, and the California Community Foundation. Designed evaluation instruments and advised teams on evaluation design and statistics.

EDUCATION

Azusa Pacific University

M.S., Research Psychology & Data Analytics *Azusa, CA*

Azusa Pacific University

B.A., Psychology (Emphasis in Quantitative Sciences) *Azusa, CA*

CERTIFICATIONS, SKILLS & TECHNOLOGIES

- Certifications: Alteryx Designer Core; JMP/SAS Joint Certificate in Research Methods
- Data & Engineering: Python; SQL; REST APIs; Alteryx; BigQuery; Tableau; Contentful; ClickUp; Alchemer; Salesforce; Claude Code; Codex; OpenAI, Google Gemini and Anthropic Claude APIs
- Research & Evaluation: study and survey design; evaluation instruments; inter-rater reliability; human-baseline validation; regression and statistical analysis; adjudication and manipulation checks
- Statistics & Classical ML: experimental design; regression and logistic regression; decision trees; model fitting and validation; nonparametric methods; reliability and agreement statistics
- AI Systems: LLM pipeline design; agentic workflows and tool design; prompt and persona design; faithfulness auditing
- Delivery: discovery, scoping and requirements; process mapping; data migration; change adoption and user enablement; stakeholder management; governance, privacy and security